Modal and Global Constraints for Phonology and Syntax

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Stony Brook University

Subregular Workshop
Rutgers University
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Outline

- Define Modal constraints
- Establish relationships between Global and Modal constraints
  How do their complexities relate?
- Characterize phonotactic patterns Modally
- Excursus on a typology of primary stress assignment

Sibilant Harmony done Globally

- **Description:**
  In Navajo, all sibilants agree in anteriority.

[si-ʔă]	‘a round object lies’
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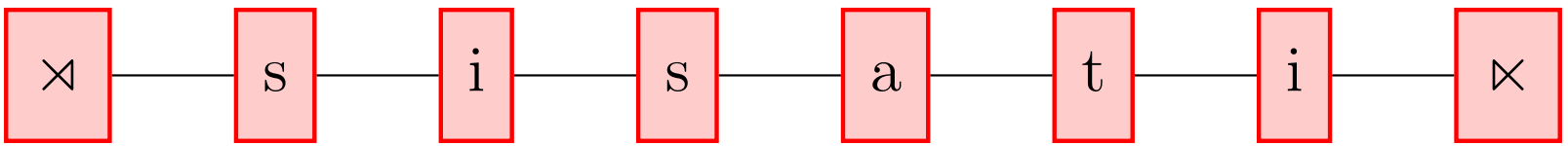
⋊ —— s —— i —— s —— a —— t —— i —— ⋉

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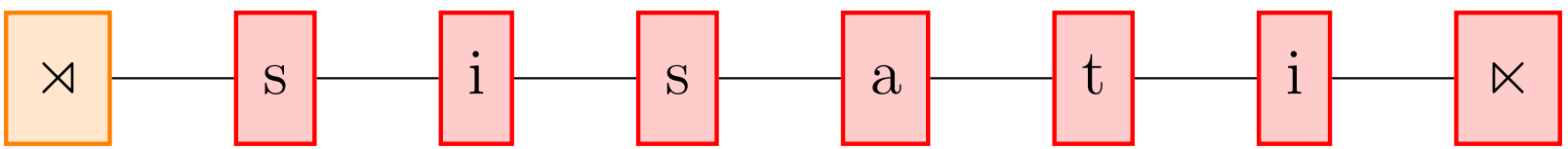
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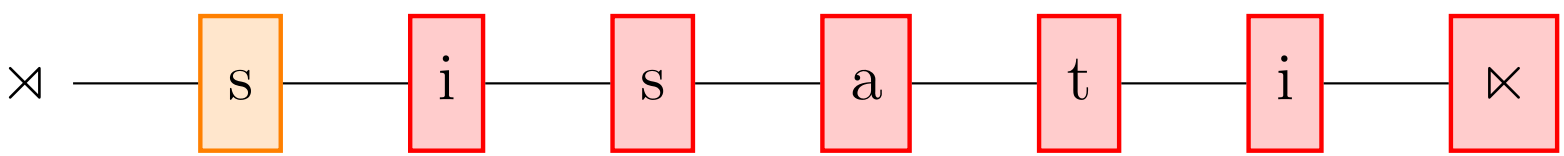


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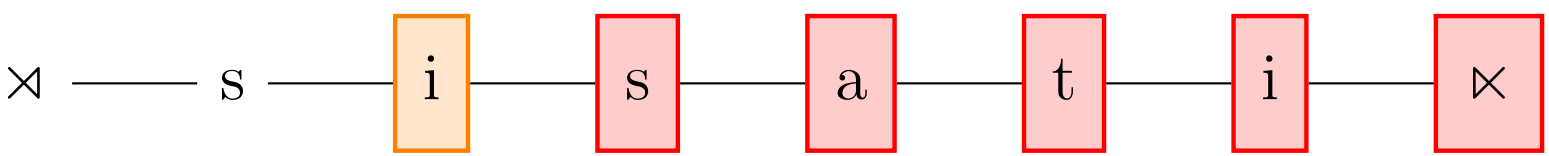


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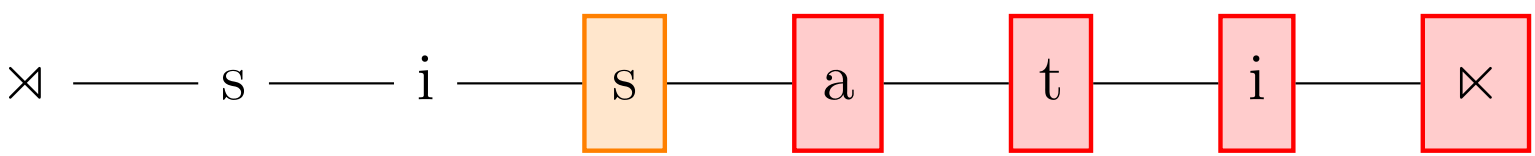


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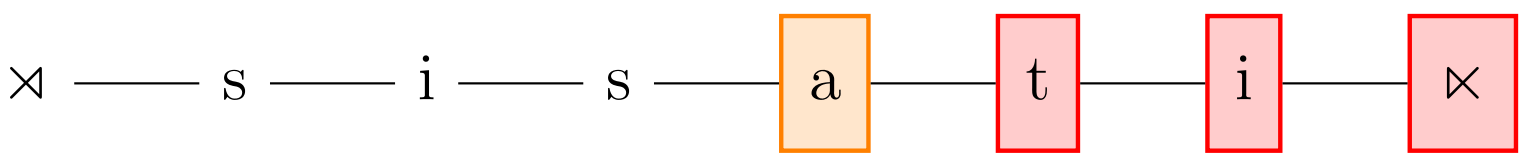


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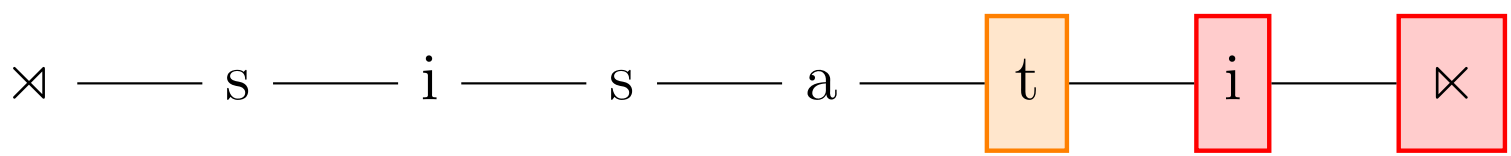


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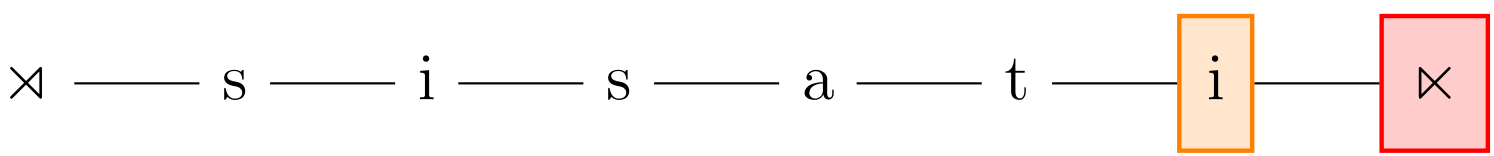


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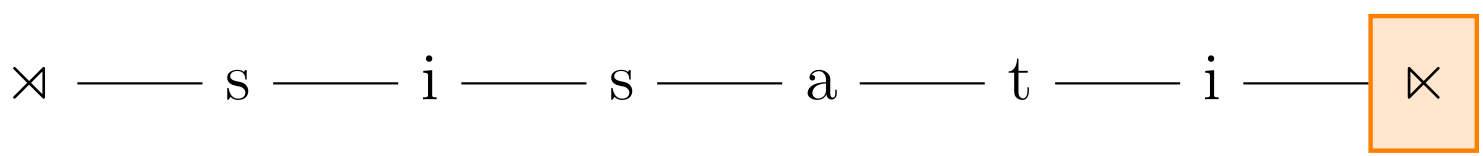


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- $\Sigma = \{a, b, c\} \rightarrow \{G_a, G_b, G_c\}$
- A Modal grammar accepts a string $s_1, s_2, ..., s_n$ iff:

  $s_1, s_2, ..., s_n \models G_{s_1}$

  $s_2, ..., s_n \models G_{s_2}$

  ...

  $s_n \models G_{s_n}$

A Modal Approach to Movement

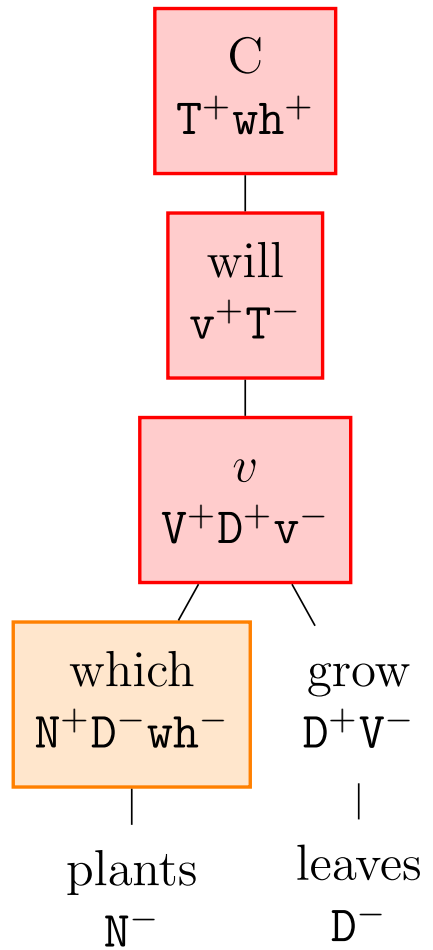
- **The Ceiling Constraint**
  Every $\mathtt{f}^-$ needs an $\mathtt{f}^+$.
- $G_{\mathtt{wh}} = \mathtt{wh}^+\Sigma^*$

Definition: a[ncestor]-string

The **a-string** of a node **n** consists of all nodes that are derivationally ordered after **n**.

This is **Tier-based Reverse 1-Definite**.

Which plants will grow leaves?



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Can modal grammars tell us about typology?
Are some phenomena better described modally?

Phonotactics: the testing ground

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Phonotactic constraints cover a wider variety of complexity classes than syntactic ones, including:

- Strictly Piecewise (SP)
- Piecewise Testable (PT)
- Locally Testable (LT)
- Multitier Generalized Definite (MTGD)

Every SP-2 Language is Modally SL-1

Global SP-2 → Modal SL-1

For each piecewise 2-factor $a...b$, add b to the set of permitted 1-factors for G_a.

Sibilant Harmony

- Global: $G = \{\int...\int, s...s\}$
- Modal:
  - $G_s = (-\int)^*$ (SL-1).
  - $G_\int = (-\mathrm{s})^*$ (SL-1).

Luganda High Tone Plateauing

In Luganda, only one span of high tones may appear within a word, and every word must end in a low tone (Jardine, 2016; Lambert, 2026).

a.	$(\ell\ell\ell)$	/ki-tabo/	→ [kitàbò]	‘book’
b.	$(\ell h\ell)$	/ki-kópo/	→ [kikópò]	‘cup’
c.	$(hhh\ell)$	/bá-ki-láb-a/	→ [bákíĺábà]	‘they see it’
d.	$(\ell\ell h\ell)$	/ki-sikí́/	→ [kisikî]	‘log’

Luganda High Tone Plateauing

- $L = l^*h^*l^+$
- Global:
  - $G = \neg h..l..h \wedge (h \rightarrow h..l)$ - PT-3 (Jardine, 2016; Lambert, 2026).
- Modal:
  - $G_h = h^*l^+$: (**SL-2!**)
  - $G_l = \Sigma^*$: (Trivial).

NOTE

An SL-2 modal grammar can represent a PT-3 grammar.

Converting Global (T)SL to Modal (T)RDef

Mdl-(T)RDef → Gbl-(T)SL

For each permitted prefix t of G_s, add st to G.

Gbl-(T)SL → Mdl-(T)RDef

For each k-gram st, where $|s| = 1$ and $|t| < k$ add t to G_s.

NOTE

Modal (T)RDef-k can accept the (T)SL-(k+1) languages.

Turkish Backness Harmony

Suffix vowels agree in backness with the closest vowel; stem vowels can be disharmonic.

a.	[ip-ler-in]	‘rope-PL-GEN’
b.	[el-ler-in]	‘hand-PL-GEN’
c.	[pul-lar-ɯn]	‘stamp-PL-GEN’
d.	[son-lar-ɯn]	‘end-PL-GEN’

Turkish Backness Harmony

- Global:
  - $G^- = \{V_f \cdot S_b | V_b \cdot S_f | S_f \cdot S_b | S_b \cdot S_f\}$
  - $\tau = \{V_f, V_b, S_f, S_b\}$
- Modal (t=1, k=1):
  - $G_{V_f} = S_b \cdot \Sigma^*$ (TRDef-1)
  - $G_{V_b} = S_f \cdot \Sigma^*$ (TRDef-1)
  - $G_{S_f} = S_b \cdot \Sigma^*$ (TRDef-1)
  - $G_{S_b} = S_f \cdot \Sigma^*$ (TRDef-1)

- V_f : front stem vowels
- V_b : back stem vowels
- S_f : front suffix vowels
- S_b : back vowels

(X-most Heavy else X-most) Stress

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Leftmost Heavy else Leftmost

- **Global:** MTRDef and PT
- **Modal:**
  - $G_{l'} = l^*$ SL-1
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A Modal Typology of XHOX

Constraints

	$\emptyset$	l^*	$(h\|l)^*$	$(-l')^*$	Σ^*
RHOR	l'	h'		h	l
LHOR	l'		h', h		l
RHOL		l', h'		h, l	
LHOL		l'	h', h	l	

$$(l' > h') > (h > l)$$

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Word Final	l', h'				h, l
RHOR	l'	h'		h	l
LHOR	l'		h', h		l
RHOL		l', h'		h, l	
LHOL		l'	h', h	l	
Word Initial			l', h', h, l		

$$(l' > h') > (h > l)$$

Other Patterns Covered

Pattern	**Global Complexity**	**Modal Complexity**
Prinimi	PT-2	MTSL-2
Culminativity	2-TDef	SL-1
Tsuut'ina Asysmmetric Harmony	TSL-2	TRDef-1
Arigibi Tone	TCo-Fin	SL-1
Kagoshima Japanese Tone	2-MTDef	MTGD
Polish Alternating Stress	SL-2	1-RDef
Abkhaz Dominant Stress	PT-3	SL-2
Korean Vowel Harmony	MTRDef-2	?MTRDef-2
Icy Targets	TSL-3	TRDef-2
$\Sigma^* ab \Sigma^* ab \Sigma^*$	LTT	LTT
$ab(ab)^+$	LT	MTSL

Summary of Complexity Findings

Global Class	**Modal Class**	**Example**
(T)SL-k	(T)RDef-(k-1)	Harmony, agreement, etc.
SP-2	SL-1	Sibilant harmony
PT-3	SL-2	Tone Plateauing
MTGD/PT-2	SL-1	Stress Assignment

Questions for the group

- How do we more clearly establish connections between the modal and global constraints?
- What stringset does Modal SL/TSL generate?
- Can we escape the propositional realm with TSL?
- Examples of languages (formal or natural) that are LT or LTT but not PT?
- What are some other typologies that we could analyze with Modal TSL?

Thank You!